We re-implemented four canonical deep RL algorithms from scratch in PyTorch and benchmarked them on four classic-control Gymnasium environments with three seeds each. The algorithms span the two main axes of the deep RL taxonomy:

- value-based vs. policy-based,
- on-policy vs. off-policy.

We report bootstrap 95% confidence intervals over seeds, pre-register four ablations, and document the implementation details that distinguish a paper recipe from a working agent.

DQN [?] (value-based, off-policy, discrete):

$$\mathcal{L}_{\text{DQN}} = \mathbb{E}\Big[\ell_{\text{Huber}}\Big(Q_\theta(s,a) - \big(r + \gamma\,(1-d)\,\max_{a'} Q_{\bar\theta}(s',a')\big)\Big)\Big]\,.$$

ε-greedy exploration, target network synced every τ_{sync} steps, Huber loss for robustness.

PPO [?] (policy-based, on-policy, discrete + continuous):

$$\mathcal{L}_{\text{PPO}} = -\,\mathbb{E}\Big[\min\big(r_t(\theta)\hat{A}_t,\ \text{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_t\big)\Big]\,.$$

Clipped surrogate as a first-order trust-region; GAE advantages with (γ, λ).

SAC [?] (policy-based, off-policy, continuous):

$$J_{\text{SAC}}(\pi) = \mathbb{E}_\pi\big[\textstyle\sum_t \gamma^t\big(r(s_t,a_t) + \alpha\,\mathcal{H}[\pi(\cdot|s_t)]\big)\big]\,.$$

Squashed-Gaussian actor with tanh-Jacobian log-prob correction; auto-tuned α via $\log\alpha$.

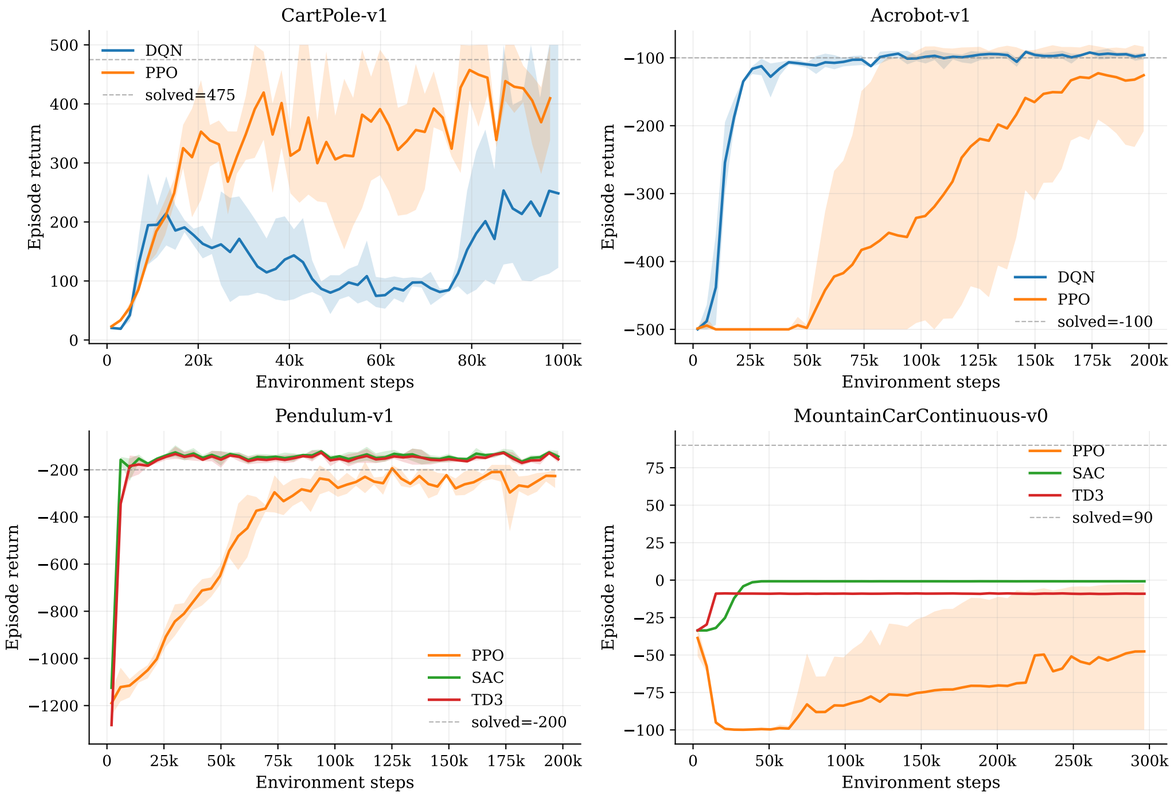
TD3 [?] (policy-based, off-policy, continuous):

$$y = r + \gamma\,(1-d)\,\min_{i\in\{1,2\}}\,Q_{\bar\phi_i}\big(s',\ \text{clip}(\pi_{\bar\theta}(s') + \tilde\epsilon,\ a_{\min}, a_{\max})\big).$$

Twin Q-targets, target policy smoothing ($\tilde\sigma$=0.2), delayed actor update every 2 critic steps.

Env	Action	Reward / budget
CartPole-v1	Discrete(2)	dense $+1$/step, 100k steps
Acrobot-v1	Discrete(3)	dense -1/step, 200k steps
Pendulum-v1	Box$[-2, 2]$	dense quadratic, 200k steps
MountainCarContinuous-v0	Box$[-1, 1]$	sparse $+100$ at goal, 300k steps

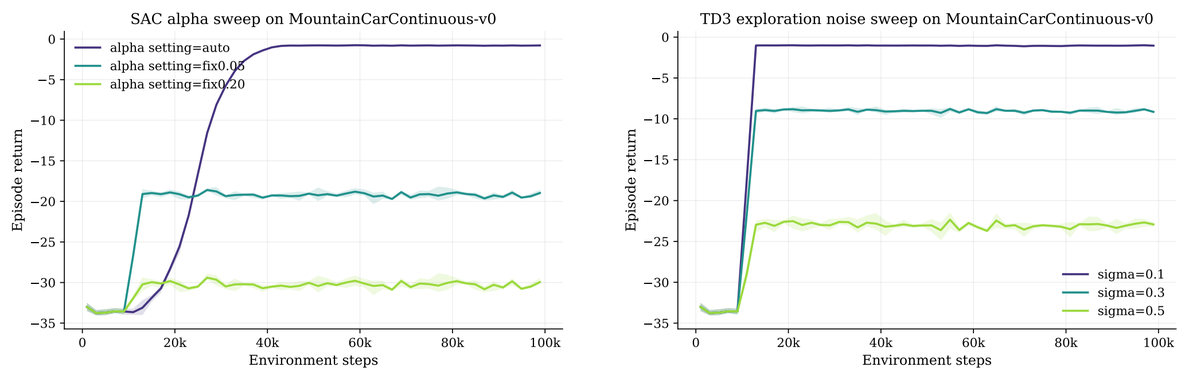
Protocol. 3 seeds $\{0, 1, 2\}$ per (algo, env). 95% bootstrap CI over seeds at every step bin. Networks: $(64, 64)$ for discrete (DQN, PPO), $(256, 256)$ for continuous (SAC, TD3), matching canonical papers. Hardware: RTX 3070 Laptop and EPFL gnoto V100. Code: PyTorch from scratch, no `stable-baselines3`.



Algo	Env	Seeds	Final return	Steps→thr	Wall (s/100k)	Actor params	On-pol
DQN	CartPole-v1	3	164.4 $[111, 259]$	100k	98	4.6k	no
DQN	Acrobot-v1	3	-96.1 $[-101, -94]$	30k	120	4.8k	no
PPO	CartPole-v1	3	380.1 $[350, 402]$	n/r	442	4.6k	yes
PPO	Acrobot-v1	3	-127.4 $[-216, -82]$	110k	457	4.8k	yes
PPO	Pendulum-v1	3	-247.9 $[-265, -226]$	95k	450	4.5k	yes
PPO	MountainCarContinuous-v0	3	-56.2 $[-100, -5]$	n/r	1961	4.4k	yes
SAC	Pendulum-v1	3	-146.1 $[-152, -137]$	7k	1183	67.3k	no
SAC	MountainCarContinuous-v0	3	-0.8 $[-1, -1]$	n/r	2713	67.1k	no
TD3	Pendulum-v1	3	-154.9 $[-160, -148]$	9k	483	67.1k	no
TD3	MountainCarContinuous-v0	3	-9.1 $[-9, -9]$	n/r	862	66.8k	no

- **CartPole.** DQN bimodal (1/3 seeds solve); PPO reliable (380, CI [350, 402]).
- **Acrobot.** DQN solves in ∼30k steps; PPO slower, one seed collapses.
- **Pendulum.** SAC and TD3 are $\sim 13\times$ more sample-efficient than PPO (SAC reaches threshold in 7.4k env steps vs PPO 95k).
- **MountainCarContinuous.** Only PPO ever discovers the +100 goal (1/3 seeds). SAC (-0.8) and TD3 (-9.1) plateau at a lazy local optimum.

Two pre-registered ablations on MountainCarContinuous test whether the available exploration knobs rescue SAC or TD3. Neither does.



SAC temperature sweep. Auto, 0.05, 0.20.
Three failure modes, 0 goal discoveries / 9 seeds.

TD3 noise sweep. $\sigma \in \{0.1, 0.3, 0.5\}$.
More noise is worse, not better.

Mechanism. Both off-policy mechanisms (entropy regularization, additive Gaussian noise) drive uniform coverage of the *local* action set, not directed exploration in state space. PPO's on-policy rollouts occasionally produce a long correlated action sequence that crosses the hill; off-policy replay samples are decorrelated and cannot.

Other ablations confirm the diary. DQN target-sync sweep is a U-shape ($\tau \in \{100, 1000, 5000\}$), explaining CartPole bimodality: τ=1000 sits at the edge of stability. PPO clip ratio: 0.1 is too conservative; 0.2 and 0.3 are tied.

- **Most expensive per iteration:** SAC (1183 to 2713 s per 100k steps) > TD3 > PPO > DQN (98 to 120 s).
- **Most compact policy:** DQN & PPO (∼4.6k actor params) vs SAC & TD3 (∼67k). Caveat: this gap follows our $(64, 64)$ vs $(256, 256)$ width choice, not the algorithms.
- **Best on continuous actions:** no single winner. SAC/TD3 dominate dense reward (Pendulum); PPO is the only one to solve sparse reward (MCC).
- **Best off-policy data efficiency:** SAC, by $12.9\times$ over PPO on Pendulum (7.4k vs 95k steps to threshold).